THESIS DRAFT

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Abstract.

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1. Introduction

Algebraic topology seeks to study topological spaces by means of *algebraic invariants*, i.e. ways of assigning to each space an algebraic object, so that two "similar" spaces receive the same object. Of these invariants, we must strike a balance between

1.1. **Main theorem.** Let X be a path-connected topological space and choose basepoints $a, b \in X$. Let $\pi_X(a, b)$ be the set of homotopy classes of paths in X from a to b. In particular when $a = b$ we obtain the fundamental group of X. Notice that $\pi_1(X, a)$ acts on $\pi_X(a, b)$ on the right by concatenation of paths, so that the free abelian group $\mathbf{Z}\pi_X(a, b)$ is a right $\mathbf{Z}\pi_1(X)$-module. Let $\mathcal{I}_a \subset \mathbf{Z}\pi_1(X, a)$ be the augmentation ideal.

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Consider now the following subspaces of X^n

$$X(n)^a := \{(x_1, \dots, x_n) \in X^n : x_1 = a \text{ or } x_i = x_{i+1} \text{ for } i \in \{1, \dots, n-1\}\}$$
$$X(n)_b := \{(x_1, \dots, x_n) \in X^n : x_n = b \text{ or } x_i = x_{i+1} \text{ for } i \in \{1, \dots, n-1\}\}$$

and let $X(n)^a_b := X(n)^a \cup X(n)_b$.

Theorem 1.1. *There is a map*

$$\mathbf{Z}\pi_X(a,b)/\mathbf{Z}\pi_X(a,b)\mathcal{I}_a^{n+1} \to H_n(X^n, X(n)^a_b; \mathbf{Z})$$

which is a surjection with kernel $\mathbf{Z}1_a$ *if* $a = b$ *and an isomorphism if* $a \neq b$.

The author first encountered this theorem in a paper of Looijenga ([Loo25], Theorem 1.1), who attributes this theorem (or a dual version with rational coefficients) to Deligne and Goncharov, ([DG05], Proposition 3.4), who attribute it to Beilinson.

1.2. Prerequisites, conventions, overview.

2. Coalgebra crash course

As suggested by the name, a *coalgebra* is the dual notion to that of an algebra. We could leave it at that, but they are much more unfamiliar objects; the maps don't go in the way we are used to. Moreover, the two objects are not dual on the nose: while the dual of every coalgebra is an algebra, the converse is not true without some finiteness assumptions. The reader should feel free to skip this chapter if they are familiar with the theory of (co)associative (co)algebras.Let R be a commutative ring.

Definition 2.1. A *coassociative R-coalgebra* is an R-module C with an R-linear map $\Delta : C \to C \otimes C$, called the *coproduct*, such that the following diagram commutes:

$$\begin{tikzcd} C \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & C \otimes C \arrow[d, "\mathrm{id} \otimes \Delta"] \\ C \otimes C \arrow[r, "\Delta \otimes \mathrm{id}"'] & C \otimes C \otimes C \end{tikzcd}$$

One way to think about the coproduct is that it tells one how to decompose a given element in the module. Consider the following example:

Example 2.2 ([JR79], §2)**.** Let $C = R[x]$, the ring of polynomials over R. Define a coproduct on a basis element x^n via

$$\Delta(x^n) = \sum_{k=0}^{n} \binom{n}{k} x^k \otimes x^{n-k}.$$

This looks awfully like the binomial formula. Indeed this coproduct tells us all the ways to decompose a set of size n into two subsets.

I learned of this example from this MathOverflow comment of Qiaochu Yuan [Yua]. Though this example is combinatorial in nature, the notion of coproduct as decomposition will come up again, when we discuss how the subdivision of topological simplices induces a coproduct on singular chains.

Given an associative algebra A it is intuitive how to compose the product to obtain a map $A^{\otimes n} \to A$. With a coassociative coalgebra (C, Δ) there is an analogous notion. Define the *iterated coproduct* $\Delta^n : C \to C^{\otimes n+1}$ inductively with $\Delta^0 = \mathrm{id}$, $\Delta^1 = \Delta$, and

$$\Delta^n = \underbrace{\Delta \otimes \mathrm{id} \otimes \cdots \otimes \mathrm{id}}_{n \text{ operations}} \circ \Delta^{n-1}.$$

Coassociativity says that we could have inserted the coproduct anywhere within the above tensor product of operations. A *morphism of coassociative coalgebras* $f : C \to D$ is an R-linear map commuting with the coproduct, i.e.

$$(f \otimes f) \circ \Delta_C = \Delta_D \circ f.$$

Just as a unital associative R-algebra A is one admitting a unital morphism $R \to A$, we say dually that a coassociative R-coalgebra C is *counital* if there is a morphism $\epsilon : C \to R$ such that the following diagram commutes:

$$\begin{tikzcd} & C \arrow[dl, "\cong"'] \arrow[d, "\Delta"] \arrow[dr, "\cong"] & \\ R \otimes C & C \otimes C \arrow[l, "\epsilon \otimes \mathrm{id}"] \arrow[r, "\mathrm{id} \otimes \epsilon"'] & C \otimes R \end{tikzcd}.$$

The simplest example of a counital coassociative R-coalgebra is R itself, with the coproduct given by $1 \mapsto 1 \otimes 1$ and the counit given by $1 \mapsto 1$. We call a morphism $\eta : R \to C$ a *coaugmentation*, and in this case say that C is *coaugmented*. Beccause η is a morphism of coalgebras, it must commute with the counit maps, so we obtain that $\epsilon_C \circ \eta_C = \epsilon_R = \mathrm{id}_R$. It then follows that $C \cong \ker \epsilon \oplus R$, and we denote this kernel by $\overline{C}$ and call it the *reduced coalgebra*. We can think of $\overline{C}$ as either a submodule or a quotient of C. The reduced coalgebra also has a coproduct $\overline{\Delta}$ given by

$$\overline{\Delta}(x) = \Delta(x) - 1 \otimes x - x \otimes 1.$$

2.1. **Conilpotency, cofreeness, and the tensor coalgebra.** Let (C, Δ) be a coaugmented coalgebra. Define the *coradical* (sometimes also called *canonical*) filtration on C as follows:

$$F_0 C = R, \quad F_r C = \{x \in \overline{C} : \bar{\Delta}^n(x) = 0 \text{ for } n \geq r\} \text{ for } r \geq 1.$$

Then we say C is *conilpotent* if this filtration is exhaustive. Conilpotency is important is in the following definition as well as later in the thesis.

Definition 2.3. The *cofree* coassociative R-coalgebra over a R-module M is a conilpotent coassociative coalgebra $\mathcal{F}^c M$ equipped with an R-linear map $s : \mathcal{F}^c M \to M$ sending 1 to 0 and satisfying the following universal property:

> *Any R-linear map $f : B \to M$ factors through $\mathcal{F}^c M$, i.e. there exists a unique map $\tilde{f} : B \to \mathcal{F}^c M$ such that $s \circ \tilde{f} = f$.*

As with other objects defined via universal properties, the cofree coalgebra is unique up to unique isomorphism. In the categorial language, this functor

$$\mathcal{F}^c : \mathsf{Mod}_R \to \mathsf{conilCoalg}_R$$

is a right adjoint to the forgetful function sending a conilpotent coalgebra to its underlying module. The reason we restrict ourselves to conilpotent coalgebras here is that the cofree objects are familiar, as we will soon see. In the general category of coalgebras they are large and unwieldy.

Definition 2.4. Let M be an R-module. The *tensor coalgebra over M*, denoted $T^c M$, is the coalgebra whose underlying module is

$$T^c M := R \oplus M \oplus M^{\otimes 2} \oplus \cdots$$

and whose coproduct $T^c M \to T^c M \otimes T^c M$ is given by

$$1 \mapsto 1 \otimes 1 \quad \text{and} \quad x_1 \otimes \cdots \otimes x_n \mapsto \sum_{i=0}^{n} (x_1 \otimes \cdots \otimes x_i) \otimes (x_{i+1} \otimes \cdots \otimes x_n).$$

For example, $x \in M$ gets mapped to $1 \otimes x + x \otimes 1$. We leave it to the reader to check that $T^c M$ is coassociative, counital, and conilpotent, with the hint that the counit is the map $T^c M \to R$ which is the identity on R and zero on higher summands.

Remark 2.5. For an R-module M, the same tensor module $R \oplus M \oplus M^{\otimes 2} \oplus \cdots$ also has an dg algebra structure given by tensor multiplication. We will often just denote $T^c M$ by TM. With this product it becomes the free associative R-algebra over M. In fact, the product and coproducts are compatible, bestowing TM with even more strucrure.

Proposition 2.6. *$T^c M$ is the cofree conilpotent coalgebra over M.*

Proof. Proposition 1.2.7, [LV12]. □

2.2. **Differential graded coalgebras.** A *differential graded* (often abbreviated to *dg*) coalgebra is a coassociative coalgebra (C, Δ) with

(1) a grading, i.e. $C = \oplus_{n \in \mathbf{N}} C_n$ such that the product sends C_n to $\oplus_{i+j=n} C_i \otimes C_j$, and
(2) a differential d sending C_i to C_{i-1}, satisfying the dual identity

$$\Delta \circ d = (d \otimes \mathrm{id} + \mathrm{id} \otimes d) \circ \Delta.$$

This second identity is the Leibniz rule; if a coalgebra endomorphism satisfies this, we call it a *coderivation*. So all differentials are coderivations. One can check that these definitions imply that the coproduct is a morphism of chain complexes. So we can think of dga coalgebras as either (1) chain complexes with a compatible coproduct, or (2) coalgebras with compatible differential. Given a dg R-coalgebra $C = \oplus_{n \in \mathbf{N}} C_n$, we can define its *suspension* sC to be C shifted up a degree, i.e. $(sV)_i = V_{i-1}$, and its *desuspension* $s^{-1}C$ to be C shifted down a degree, i.e. $(s^{-1}C)_j = C_{j+1}$. Then any $x \in C_n$ determines an element $sx \in (sC)_{n+1}$ and $s^{-1}x \in (s^{-1}C)_{n-1}$. Defining these turns out to be helpful for degree bookkeeping reasons.

The following lemma helps us simplify many verifications. It tells us that to understand derivations of a free algebra, or coderivations of a cofree coalgebra, it suffices to understand them on one component.

Lemma 2.7. *Let M be an R-module. Any linear map $M \to TM$ extends uniquely to a derivation $TM \to TM$. Dually, any coderivation $T^c M \to T^c M$ is uniquely determined by the projection $T^c M \to T^c M \to M$.*

Proof. These follow from the more general Proposition 1.1.8 and 1.2.9 in [LV12]. □

Example 2.8. Consider the singular chains $(C_\bullet(X; \mathbf{Z}), \partial)$ on a topological space X with basepoint x. They form a coaugmented dg $\mathbf{Z}$-coalgebra. For the coproduct, notice that we always have a diagonal map $x \mapsto (x, x)$, which induces a chain map $C_\bullet(X) \to C_\bullet(X \times X)$. Now there a natural quasi-isomorphism $C_\bullet(X) \otimes C_\bullet(X)$, given by the decomposition of the topological simplices. Because decomposing a simplex reduces their dimension, we get conilpotence. $C_\bullet(X)$ is counital with the map $\epsilon : C_\bullet(X) \to \mathbf{Z}$ given by sending all 0-simplicies to 1 and has a coaugmentation $\eta : \mathbf{Z} \to C_\bullet(X)$ which sends 1 to the 0-simplex at x. In Section [], we give an explicit definition of this coproduct (the *Alexander–Whitney* map).

2.3. **Comodules.** Continuing with our previous example of singular chains on a space X, the situation changes if we want to talk about two distinct basepoints $a, b \in X$. In this case there is no preferred choice of coaugmentation for the coalgebra $C_\bullet(X)$. For this we need the notion of a comodule.

Definition 2.9. A *left comodule* over a unital R-coalgebra (C, Δ, ϵ) is an R-module N with a left action

$$\Delta^L : N \to C \otimes N$$

compatible with the coproduct and counit in the sense that the following diagrams commute:

$$\begin{tikzcd} N \arrow[r, "\Delta^L"] \arrow[d, "\Delta^L"'] & C \otimes N \arrow[d, "\mathrm{id} \otimes \Delta^L"] \\ C \otimes N \arrow[r, "\Delta \otimes \mathrm{id}"'] & C \otimes C \otimes N \end{tikzcd} \qquad \begin{tikzcd} N \arrow[d, "\Delta^L"'] \arrow[dr, "\cong"] & \\ C \otimes N \arrow[r, "\epsilon \otimes \mathrm{id}"'] & R \otimes N \end{tikzcd}$$

We can similarly define a *right* comodule over C as an R-module M with the analogous right action

$$\Delta^R : M \to M \otimes C.$$

If a left and right comodule M also satsifies the following compatibility condition:

$$\begin{tikzcd} M \arrow[r, "\Delta^L"] \arrow[d, "\Delta^R"'] & C \otimes M \arrow[d, "\mathrm{id} \otimes \Delta^R"] \\ M \otimes C \arrow[r, "\Delta^L \otimes \mathrm{id}"'] & M \otimes C \otimes M \end{tikzcd}$$

then we call M a *co-bimodule*. The simplest example of a co-bimodule over a coalgebra C is C itself, where the left right actions are just given by the coproduct. Returning to the example of singular chains, let $C = C_\bullet(X)$ and let a be a basepoint. Then $C_\bullet(\{a\})$ is a bi-comodule over $C_\bullet(X)$ via the left and right coactions

$$C_\bullet(\{a\}) \xrightarrow{\Delta} C_\bullet(\{a\}) \otimes C_\bullet(\{a\}) \xrightarrow{\iota_a \otimes \mathrm{id}} C_\bullet(X) \otimes C_\bullet(\{a\})$$

$$C_\bullet(\{a\}) \xrightarrow{\Delta} C_\bullet(\{a\}) \otimes C_\bullet(\{a\}) \xrightarrow{\mathrm{id} \otimes \iota_a} C_\bullet(X) \otimes C_\bullet(\{a\}).$$

If $a \neq b$, we can take one of $C_\bullet(\{a\})$ and $C_\bullet(\{b\})$ to be a left comodule and the other to be a right comodule over $C_\bullet(X)$. The usefulness of these definitions will hopefully be clear as this thesis progresses.

3. (Co)simplicial methods

> *A cosimplicial object of a category $\mathcal{C}$ could be defined simply as a simplicial object of the opposite category $\mathcal{C}^{\mathrm{op}}$. This is not really how the human brain works...*
>
> —Stacks Project, [Sta26, Tag 016I]

In this chapter we define cosimplicial objects, the totalization of a cosimplicial space, and provide some examples. We will state the Dold–Kan correspondence and use it to prove a useful statement for later on. Finally we conclude with an exposition of the Alexander–Whitney map, which couldn't find a home in any other chapter.

The *simplex category*, denoted Δ, is the category with

(1) objects: finite nonempty totally ordered sets. We write $[n]$ for the set $\{0 < 1 < \cdots < n\}$,
(2) morphisms: order-preserving maps, i.e. if $i \leq j$ then $f(i) \leq f(j)$.

Then a *cosimplicial object* in a category $\mathcal{C}$ is a functor $X^\bullet : \Delta \to \mathcal{C}$. We denote the image of $[n]$ under $X^\bullet$ by X^n. A morphism of two simplicial objects $X^\bullet, Y^\bullet : \Delta \to \mathcal{C}$ is a natural transformation of functors, i.e. morphisms $X^n \to Y^n$ for all n that commute with morphisms in Δ. We will denote the category of of cosimplicial objects in $\mathcal{C}$ by $\mathsf{cs}\mathcal{C}$. In this thesis we will concern ourselves with the cases where $\mathcal{C} = \mathsf{Top}$ or $\mathcal{C} = \mathsf{Set}$.

Remark 3.1. Even though simplicial objects are contravariant functors, the convention is to denote the simplicies using subscripts, e.g. X_n. Conversely, even though simplicial spaces are covariant functors, the convention is to denote the cosimplicies with superscripts. Maybe the presence of the suffix co- explains this.

Recall that the morphisms in the simplex category are generated by two distinguished classes of maps. For $n \geq 1$ and $j \in [n]$, we have injections $\delta_j : [n-1] \to [n]$ where δ_j skips $j \in [n]$. For $n \geq 0$ and $j \in [n]$, we have $n+1$ surjections $\sigma_j : [n+1] \to [n]$ where σ_j sends both $j, j+1$ to $j \in [n]$. For a cosimplicial object $X^\bullet : \Delta \to \mathcal{C}$, we call the images of the δ_j *coface maps*, usually denoted d^j, and the images of the σ_j *codegeneracy maps*, usually denoted s^j. So to specify a cosimplicial object in $\mathcal{C}$ it also suffices to list a sequence of objects $X^n \in \mathcal{C}$ for $n \geq 0$, as well as coface and codegeneracy maps satisfying the following *cosimplicial identities*:

(1) If $i < j$, then $d^j \circ d^i = d^i \circ d^{j-1}$.
(2) If $i < j$, then $s^j \circ d^i = d^i \circ s_{j-1}$.
(3) $\mathrm{id} = s^j \circ d^j = s^j \circ d^{j+1}$.
(4) If $i > j+1$, then $s^j \circ d^i = d^{i-1} \circ s^j$.
(5) If $i \leq j$, then $s^j \circ s^i = s^i \circ s^{j+1}$.

One should think of a cosimplicial object $X^\bullet : \Delta \to \mathcal{C}$ as a diagram

$$X^0 \;\substack{\longrightarrow\\ \longleftarrow\\ \longrightarrow}\; X^1 \;\substack{\longrightarrow\\ \longleftarrow\\ \longrightarrow\\ \longleftarrow\\ \longrightarrow}\; X^2 \;\substack{\longrightarrow\\ \longleftarrow\\ \longrightarrow\\ \longleftarrow\\ \longrightarrow\\ \longleftarrow\\ \longrightarrow}\; \cdots$$

where the rightward pointing arrows are the coface maps and the leftward pointing arrows are the codegeneracy maps. In general it helps to think of the coface maps as "duplicating a coordinate" and the codegeneracy maps as "forgetting a coordinate." We will see this in the following examples.

Example 3.2 (Topological simplicies)**.** Define the functor $\Delta^\bullet : \Delta \to \mathsf{Top}$ which sends $[n]$ to the topological n-simplex:

$$\Delta^n = \{(x_1, \ldots, x_n) \in \mathbf{R}^n : 0 \leq x_1 \leq \cdots \leq x_n \leq 1\}.$$

Then the coface maps $\Delta^{n-1} \to \Delta^n$ are the inclusions of faces, where d^j is the inclusion of the face opposite the jth vertex. The codegeneracy map s^j collapses the line joining the jth and jth vertex. In coordinates, we have

$$d^j(x_1, \ldots, x_n) = (x_1, \ldots, x_j, x_j, \ldots, x_{n-1}),$$
$$s^j(x_1, \ldots, x_n) = (x_1, \ldots, \widehat{x_j}, \ldots, x_n)$$

where $\widehat{\ }$ indicates omission.

Example 3.3 (Path spaces)**.** Let X be a topological space with basepoints $a, b \in X$. Define the cosimplicial space $P^\bullet_{a,b}X$ whose cosimplicies are

$$P^0_{a,b}X = \{*\}, \quad P^n_{a,b}X = X^n \text{ for } n \geq 1.$$

The coface maps $d^j : P^{n-1}_{a,b}X \to P^n_{a,b}$ are given by

$$d^j(x_1, \ldots, x_{n-1}) = \begin{cases} (a, x_1, \ldots, x_{n-1}) & j = 0 \\ (x_1, \ldots, x_j, x_j, \ldots, x_{n-1}) & j \in \{1, \ldots, n-1\} \\ (x_1, \ldots, x_{n-1}, b) & j = n \end{cases}$$

The codegeneracy maps $s^j : P^{n+1}_{a,b}X \to P^n_{a,b}X$ are given by

$$s^j(x_1, \ldots, x_{n+1}) = (x_1, \ldots, \widehat{x_{j+1}}, \ldots x_{n+1}), \quad j \in \{0, \ldots, n\}$$

where the $\widehat{\ }$ denotes omission.

We leave it to the reader to verify that the maps in both examples satisfy the cosimplicial identities. Why we refer to this example by *path spaces* will become evident in the following section.

3.1. **Totalization.** Cosimplicial spaces provide a useful model for many types of topological spaces, including the based path and loop spaces. This is done via *totalization*, which is dual to the notion of geometric realization of a simplicial set.

Given a cosimplicial space $X^\bullet : \Delta \to \mathsf{Top}$, define the *totalization* of $X^\bullet$ to be the space of maps from the cosimplicial simplices to $X^\bullet$:

$$\mathrm{Tot}(X^\bullet) := \mathrm{Hom}_{\mathsf{csTop}}(\Delta^\bullet, X^\bullet),$$

i.e. maps $f^n : \Delta^n \to X^n$ for all $n \geq 0$ that commute with the coface and codegeneracy maps. We topologize it as a subspace of $\prod_{n\geq 0} \mathrm{Hom}(\Delta^n, X^n)$ with the compact-open topology. So totalization gives us a functor from csTop to Top.

Returning to Example 3.3, what seems to be happening is that by iterating the face maps, we are creating finer and finer piecewise paths whos endpoints agree, and basepoints are at a and b. Indeed,

Proposition 3.4. $\mathrm{Tot}(P^\bullet_{a,b}X)$ *is homeomorphic to the path space* $\Omega_{a,b}X$ *of paths in* X *beginning at* a *and ending at* b.

Proof. A point of $\mathrm{Tot}(P^\bullet_{a,b}X) = \mathrm{hom}_{\mathsf{csTop}}(\Delta^\bullet, P^\bullet_{a,b}X)$ us a sequence of continuous maps

$$f = \{f_i : \Delta^i \to X^i\}_{i\geq 0}$$

commuting with the coface and codegeneracy maps. Fix $n \geq 2$ and $k \in \{1, \ldots, n\}$. Consider the following composition of codegeneracy maps

$$\alpha_{n,k} := \underbrace{s^{n-1} \circ s^{n-2} \circ \cdots \circ s^{k-2} \circ s^k \circ \cdots \circ s^0}_{n-1 \text{ maps}}$$

where we compose all the degeneracies except for s^{k-1}. This gives us a map $\Delta^n \to \Delta^1$ and likewise for $X^n \to X^1$. Then for $f = \{f_0, f_1, \ldots\} \in \mathrm{Tot}(P^\bullet_{a,b}X)$, we have by commutativity that

$$f_1 \circ \alpha_{n,k} = \alpha_{n+1,k} \circ f_n.$$

But now the right hand side is just picking out the kth coordinate of f_n. Hence for $n \geq 2$, f_n is completely determined by f_1, so that the projection $\Phi : \mathrm{Tot}(P^\bullet_{a,b}X) \to \mathrm{Map}(\Delta^1, X)$ given by $\{f_i\}_{i\geq 0} \mapsto f_1$ is injective.

We claim next that Φ is actually a map into $\Omega_{a,b}X \subset \mathrm{Map}(\Delta^1, X)$. The cosimplicial relations imply

$$f_1 \circ d^0 = d^0 \circ f_0 = \mathrm{const}_a, \quad f_1 \circ d^1 = d^1 \circ f_0 = \mathrm{const}_b$$

so $f_1(0) = a$ and $f_1(1) = b$ as desired.

Lastly we define an inverse to $\Phi : \mathrm{Tot}(P^\bullet_{a,b}X) \to \Omega_{a,b}X$. For a given path $\gamma \in \Omega_{a,b}X$ consider the family of maps $\{f_i : \Delta^i \to X^i\}_{i\geq 0}$ given as follows. We let f^0 be the constant map, $f^1 = \gamma$, and for $n \geq 2$ define

$$f_n(x_1, \ldots, x_n) = (\gamma(x_1), \gamma(x_1 + x_2), \ldots, \gamma(x_1 + \cdots + x_n)).$$

Clearly this is an inverse to Φ. We leave it to the reader to check that the family of maps $\{f_i\}$ commutes with the coface and codegeneracies, and that Φ and its inverse are continuous maps. □

We will call $P^\bullet_{a,b}X$ a cosimplicial model for the path space of X. When $a = b$ we get a cosimplicial model for the based loop space of X. It turns out that the cosimplicial space $P^\bullet_{x,x}X$ is in some sense the underlying cosimplicial set of the cobar construction of $C_\bullet(X)$, as we will see in the course of this thesis.

3.2. **Dold–Kan.** Recall that the standard Dold–Kan correspondence gives an equivalence of categories between simplicial abelian groups and (non-negative) chain complexes of abelian groups. We'll restate some important results and their duals.

Definition 3.5. Let A be a simplicial abelian group. Its *normalized chain complex* $(NA_\bullet, \partial)$ has

$$NA_n = \bigcap_{i=1}^{n} \ker(d^i : A_n \to A_{n-1}), \quad \partial = d_0 : NA_n \to NA_{n-1}$$

and its *Moore complex* $(MA_\bullet, \partial')$ has

$$MA_n = A_n, \quad \partial' = \sum_{i=0}^{n} (-1)^i d_i : A_n \to A_{n-1}.$$

The following results and their proofs can be found in [GJ09], Chapter III.2:

Theorem 3.6 (Dold–Kan correspondence)**.** *Let A be a simplicial abelian group. Then:*

(1) The functor $N : \mathsf{sAb} \to \mathsf{Ch}_+(\mathsf{Ab})$ is an equivalence of categories between simplicial abelian groups and non-negative chain complexes of abelian groups.
(2) The inclusion of chain complexes $NA \hookrightarrow MA$ is a chain homotopy equivalence, natural in A.
(3) There is a functorial direct sum decomposition $MA = NA \oplus DA$, where DA is the subcomplex generated by the images of all the degeneracy maps. Consequently, DA is acyclic.

Now we dualize. Recall that for simplicial objects, the face maps d_i lower degree. For cosimplicial objects, the face maps d^i raise degree, so we should get an equivalence between cosimplicial abelian groups and cochain complexes of abelian groups. This is indeed what happens.

Definition 3.7. Let $C^\bullet$ be a cosimplicial abelian group. Its normalized cochain complex $(NC^\bullet, \partial)$ has

$$NC^n = \operatorname{coker} \bigoplus_{i=1}^{n} (d^i : C^{n-1} \to C^n), \quad \partial = d^0 : NC^n \to NC^{n+1}$$

and its Moore complex has

$$MC^n = C^n, \quad \partial' = \sum_{i=0}^{n+1} (-1)^i d^i : C^n \to C^{n+1}.$$

Analogously we obtain the following statements:

Theorem 3.8 (Dold–Kan correspondence, dual)**.** *Let C be a cosimplicial abelian group.*

(1) The functor $N : \mathsf{csAb} \to \mathsf{coCh}_+(\mathsf{Ab})$ is an equivalence of categories between cosimplicial abelian groups and non-negative cochain complexes of abelian groups.
(2) The quotient of cochain complexes $MC \to NC$ is a cochain homotopy equivalence, natural in C.
(3) There is a functorial direct sum decomposition $MC = NC \oplus DC$, where DC is the subcomplex generated by the images of all the coface maps. Consequently, DC is acyclic.

There is another characterization of the normalized cochain complex given a cosimplicial abelian groups:

Lemma 3.9. *For $C^\bullet$ a cosimplicial abelian group, its normalized chain complex is isomorphic to the complex $KC^\bullet$ with*

$$KC^n = \bigcap_{i=1}^{n-1} \ker(s^i : C^n \to C^{n-1})$$

and differential the restriction of the alternating sum of coface maps.

Proof. The proof is dual to the discussion in [GJ09] following the proof Corollary 2.3. □

Really we could have stated the previous four theorems for simplicial objects in any abelian category. In the following discussion, we replace "abelian group" with "R-module" for some commutative ring R. Consider the free R-module functor $R[-] : \mathsf{Set} \to \mathsf{Mod}_R$. Then given any cosimplicial set $X^\bullet : \Delta \to \mathsf{Set}$, the composition of functors $R[X^\bullet]$ yields a cosimplicial R-module. We first state a useful lemma, which was taken by a comment of Tom Goodwillie on a MathOverflow thread [Goo] related to the upcoming Proposition 3.11.

Lemma 3.10 (Goodwillie's lemma)**.** *Let $n > 0$ and choose $x, y \in X^n$. If x, y are not in the image of any coface maps $d^j : X^{n-1} \to X^n$, then $d^i x = d^j y$ implies $i = j$ and $x = y$.*

Proof. First suppose $i < j$. Then $d^i x = d^j y$ implies

$$d^i s^{j-1} x = s^j d^i x = s^j d^j y = y$$

so that y is in the image of a coface map, a contradiction. The same argument for $j < i$ leaves only the possibility $i = j$. But $s^i d^i = \mathrm{id}$, implying that d^i is injective, so $x = y$. □

Proposition 3.11. *Under the Dold–Kan correspondence, the normalized cochain complex of $R[X^\bullet]$ has zero cohomology in positive degree.*

Proof. Let $NR[X^\bullet]$ be the normalized cochain complex of $R[X^\bullet]$. In each degree, we have

$$NR[X^n] = R[X^n]/D^n$$

where D^n is the free module generated by $\cup_{i=1}^n \operatorname{im} d^i$. It follows that $NR[X^n]$ is the free module on the set $U^n := X^n \setminus \cup_{i=1}^n \operatorname{im} d^i$. Let A_n be the subset of X^n consisting of elements not in the image of any coface map, and define

$$B^n = \{d^0 x : x \in A^{n-1}, d^0 x \notin \cup_{i=1}^n d^i\}.$$

We make the following two claims:

(1) For $n \geq 1$, $U^n = A^n \sqcup B^n$

Proof. Disjointness and the inclusion $A^n \sqcup B^n \subset U^n$ are clear. Conversely, let $x \in U^n$. If $x \notin A^n$, then $x = d^k(y)$ for some $y \in X^{n-1}$, in fact $k = 0$ by definition of U^n. It remains to show $y \in A^{n-1}$. If $y \notin A^{n-1}$ then $y = d^l(z)$ for some $z \in A^{n-2}$ (here if $n = 1$ this is impossible). So $x = d^0 d^l(z)$, and applying a cosimplicial identity we have $x = d^{l+1} d^0(z)$, contradicting the fact that $x \in U^n$. This completes the proof of the claim. □

(2) For $n \geq 2$, $B^n = \operatorname{im}(d^0 : X^{n-1} \to X^n)$.

Proof. It suffices to show that if $x \in \operatorname{im} d^0$, then $x \notin \cup_{i=1}^n \operatorname{im} d^i$. Suppose for contradiction that

$$x = d^0 a = d^j b$$

for $a \in A^{n-1}$, $b \in X^{n-1}$, and $j \geq 1$. Applying s^0 to both sides we obtain

$$a = s^0 d^0 a = s^0 d^j b = \begin{cases} b & j = 1 \\ d^{j-1} s^0 b & j > 1 \end{cases}.$$

In the first case we have $d^0 a = d^1 b$, contradicting Goodwillie's lemma since $a = b \in A^{n-1}$. In the second case we see that a is in the image of a coface map d^{j-1}, a contradiction. This completes the proof. □

Now we compute the coface map

$$d^0 : A^n \sqcup B^n \to A^{n+1} \sqcup B^{n+1}.$$

By the two claims, d^0 is a bijection $A^n \to B^{n+1}$. On B^n, we have $d^0 x = d^0 d^0 y = d^1 d^0 y$ for some $y \in X^{n-1}$ by cosimplicial identities, so this is zero in NC^n. It follows that

$$H^n(R[X^\bullet]) = \frac{\ker(d^0 : NC[X^n] \to NC[X^{n+1}])}{\operatorname{im}(d^0 : NC[X^{n-1}] \to NC[X^n])} = \frac{B^n}{B^n} = 0$$

completing the proof. □

Remark 3.12. The analogue of Proposition 3.11 for simplicial sets is completely false! For example, given a simplicial set $K_\bullet$, the chain complex $NR[K_\bullet]$ computes the homology of K. If these groups were all zero we would be in trouble.

3.3. **Alexander–Whitney and Eilenberg–Zilber.** In the proof of Looijenga's theorem there is a step where one needs to move from singular chains on a product to the product of chains. There is a canonical way to do so, called the Alexander–Whitney map, with a natural inverse called the Eilenberg–Zilber map.

The Alexander–Whitney map is a way of moving between the category of simplicial abelian groups and the category of chain complexes of abelian groups that preserves their respective monoidal structure, which we now recall. The monoidal product on sAb is given levelwise: for $A, B \in \mathsf{sAb}$, we have

$$(A \otimes B)_n = A_n \otimes B_n$$

and the face and degeneracy maps are inherited naturally. The monoidal product on the category of chain complexes is given as follows: for $X, Y \in \mathsf{Ch}_+(\mathsf{Ab})$, define

$$(X \otimes Y)_n = \bigoplus_{p+q=n} X_p \otimes Y_q.$$

and the new differential $(X \otimes Y)_n \to (X \otimes Y)_{n-1}$ by the Koszul sign rule:

$$\partial_{X \otimes Y}(x \otimes y) = \partial_X x \otimes y + (-1)^{\deg x} x \otimes \partial_Y y.$$

Definition 3.13. Let $M : \mathsf{sAb} \to \mathsf{Ch}_+(\mathsf{Ab})$ be the Moore complex functor. The *Alexander–Whitney* map $\mathcal{AW}$ is a natural transformation of the functors

$$M(- \otimes -), M(-) \otimes M(-) : \mathsf{sAb} \times \mathsf{sAb} \to \mathsf{Ch}_+(\mathsf{Ab})$$

whose components are given as follows. For $A, B \in \mathsf{sAb}$ and $a \in A_n, b \in B_n$, define the map

$$\mathcal{AW}_{A,B} : M(A \otimes B) \to M(A) \otimes M(B), \quad a \otimes b \mapsto \bigoplus_{p+q=n} \tilde{d}^p(a) \otimes d^q(b)$$

where d^p is the face map induced by $[p] \to [n], i \mapsto i$, and d^q is the face map induced by $[q] \to [n], i \mapsto i + p$.

Proposition 3.14. *$\mathcal{AW}$ restricts to a natural transformation*

$$N(- \otimes -) \implies N(-) \otimes N(-)$$

where N is the normalized chains functor. Moreover, for $A, B \in \mathsf{sAb}$,

$$\mathcal{AW}_{A,B} : N(A \otimes B) \to N(A) \otimes N(B)$$

is a natural quasi-isomorphism.

Proof. See Section 2.5.8 of Kerodon [Lur26, Tag 00S0]. □

This is a significant fact in homological algebra; now we specialize to the world of topology. Let X, Y be topological spaces, and $C_\bullet(-)$ the singular chains functor from Top to sAb. Then $MC_\bullet(X)$ is the singular chain complex of X in the usual sense.

Corollary 3.15. *There is a natural quasi isomorphism*

$$MC_\bullet(X \times Y) \to MC_\bullet(X) \otimes MC_\bullet(Y).$$

Proof. Recall that $C_\bullet(-)$ is the composition of the following functors:

$$\mathsf{Top} \xrightarrow{X \mapsto \mathrm{Map}(\Delta^\bullet, X)} \mathsf{sSet} \xrightarrow{A_\bullet \mapsto \mathbf{Z}[A_\bullet]} \mathsf{sAb}$$

and under this composition we get

$$X \times Y \mapsto \mathrm{Map}(\Delta^\bullet, X \times Y) = \mathrm{Map}(\Delta^\bullet, X) \times \mathrm{Map}(\Delta^\bullet, Y) \mapsto \mathbf{Z}[\mathrm{Map}(\Delta^\bullet, X)] \otimes \mathbf{Z}[\mathrm{Map}(\Delta^\bullet, Y)]$$

because the free abelian group functor sends Cartesian products of sets to tensor products of groups. So the simplicial abelian group $C_\bullet(X \times Y)$ is equal to $C_\bullet(X) \otimes C_\bullet(Y)$. Then, by Proposition 3.14 and the Dold–Kan correspondence we are done. □

Really what we have defined here is a topological Alexander–Whitney map which is a natural transformation of the functors $\mathsf{Top} \times \mathsf{Top} \to \mathsf{sAb}$:

$$MC_\bullet(- \times -) \to MC_\bullet(-) \otimes MC_\bullet(-)$$

For topological spaces X, Y, denote the component map by $\mathcal{AW}_{X,Y}$. From now we will use $C_\bullet(X)$ to mean both the singular (Moore) chain complex and the simplicial abelian group, depending on context.

Corollary 3.16. *For $q \geq 0$ and a topological space X, there is a natural quasi-isomorphism of chain complexes*

$$\mathcal{AW}_q : C_\bullet(X^q) \to C_\bullet(X)^{\otimes q}.$$

Proof. Iterate the topological Alexander–Whitney map

$$C_\bullet(X \times X^{q-1}) \xrightarrow{\mathcal{AW}_{X,X^{q-1}}} \cdots \xrightarrow{\mathrm{id}^{\otimes q-2} \otimes \mathcal{AW}_{X,X}} C_\bullet(X)^{\otimes q}.$$

Since each component is a natural quasi-isomorphism, so is the composition. □

As promised, there is also a map going the other way, called the *Eilenberg–Zilber* map, a natural transformation of functors

$$\mathcal{EZ} : M(-) \otimes M(-) \implies M(- \otimes -).$$

We are content without having an explicit formula for this map; it is slightly more finnicky than the formula for Alexander–Whitney, and all we need is the following theorem:

Theorem 3.17 (Eilenberg–Zilber theorem, [EZ53])**.** *For simplicial abelian groups A, B, we have chain homotopies*

$$\mathcal{AW} \circ \mathcal{EZ} \simeq \mathrm{id}_{M(A) \otimes M(B)}, \quad \mathcal{EZ} \circ \mathcal{AW} \simeq \mathrm{id}_{M(A \otimes B)}.$$

It follows taht $\mathcal{EZ}$ is also a quasi-isomorphism, and these maps are inverses. Specializing to our topological setting, we obtain

Corollary 3.18. *For $q > 0$ and a topological space X, there is a natural quasi-isomorphim of chain complexes*

$$\mathcal{EZ}_q : C_\bullet(X)^{\otimes q} \to C_\bullet(X^q).$$

Moreover, $\mathcal{EZ}_q \circ \mathcal{AW}_q$ is chain homotopic to the idenitty on $C_\bullet(X^q)$ and $\mathcal{AW}_q \circ \mathcal{EZ}_q$ is chain homotopic to the identity on $C_\bullet(X)^{\otimes q}$. □

Remark 3.19. The proof of the Eilenberg–Zilber theorem appearing in [EZ53] does not contain any explicit formulas, making use instead of acyclic models. Explicit formulas for the maps involved can be found in [EML53], stated in terms of (p,q)-*shuffles*, which encode to the way simplicies can be multiplied and divided.

4. Adams' cobar construction

In this chapter we define the cobar construction of a coalgebra and introduce Adams' remarkable 1956 theorem [Ada56], which (roughly) says that given a based topological space (X,x), one can recover the singular chains of the based loop space $\Omega_x X$ via a purely algebraic construction on the singular chains on X. We then extend the cobar construction to account for two basepoints by using comodules over a coalgebra, and conclude by elaborating on the relationship between the cobar construction and the fundamental group.

4.1. **The cobar construction.** Let R be a commutative unital ring and let $(C = \overline{C} \oplus R, \Delta)$ be a dg coaugmented counital conilpotent R-coalgebra. The cobar construction is a functor from this category of coalgebras to the category of dg associative algebras. There are many references for the following definition, including Adams' original paper [Ada56] and [Che73]. We follow the presentation given in [Riv22].

Definition 4.1. Let (C,Δ,∂) be as above. The *cobar construction* on C is the dg R-algebra whose underlying algebra is the tensor algebra over the desuspension of the reduced coalgebra C:

$$\mathsf{Cobar}(C) := T(s^{-1}\overline{C}) = R \oplus s^{-1}\overline{C} \oplus s^{-1}\overline{C}^{\otimes 2} \oplus \cdots$$

and whose differential is given by extending the linear map

$$-s^{-1} \circ \partial \circ s^{+1} + (s^{-1} \otimes s^{-1}) \circ \Delta \circ s^{+1} : s^{-1}\overline{C} \to s^{-1}\overline{C} \oplus s^{-1}\overline{C}^{\otimes 2}$$

to all of $T(s^{-1}\overline{C})$, which yields a linear map of degree -1 from $T(s^{-1}\overline{C})$ to itself.

The definition of the differential here is valid owing to Lemma 2.7, which says that it suffices to define the differential on the component $s^{-1}\overline{C} \subset T(s^{-1}\overline{C})$. Regardless, it will be helpful to be a little more explicit. By the discussion in the previous chapter we have that as a chain complex, $\mathsf{Cobar}(C)$ has terms

$$\begin{aligned}
\mathsf{Cobar}_k(C) &= (T(s^{-1}\overline{C}))_k \\
&= \bigoplus_{q\geq 0} (s^{-1}\overline{C})_k^{\otimes q} \\
&= \bigoplus_{q\geq 0} \bigoplus_{n_1+\cdots+n_q=k} (s^{-1}\overline{C})_{n_1} \otimes \cdots \otimes (s^{-1}\overline{C})_{n_q} \\
&= \bigoplus_{q\geq 0} \bigoplus_{n_1+\cdots+n_q=k} \overline{C}_{n_1+1} \otimes \cdots \otimes \overline{C}_{n_q+1} \\
&= \bigoplus_{q\geq 0} \bigoplus_{m_1+\cdots+m_q=k+q} \overline{C}_{m_1} \otimes \cdots \otimes \overline{C}_{m_q} \\
&= \bigoplus_{p+q=-k} \bigoplus_{m_1+\cdots+m_q=-p} \overline{C}_{m_1} \otimes \cdots \otimes \overline{C}_{m_q}
\end{aligned} \tag{4.2}$$

The purpose of introducing the dummy variable p at the end of this tedious indexing exercise was to suggest that we should be able to realize the cobar construction as the totalization of a second quadrant bicomplex. Indeed, in the cobar differential, we utilize both the internal differential ∂ of C with its differential graded structure as well as the coproduct Δ of C as a coalgebra.

Definition 4.3. Let (C, Δ, ∂) be as above. The *cobar bicomplex* is the bicomplex $\mathsf{Cobar}_{\bullet,\bullet}(C)$ with

$$\mathsf{Cobar}_{p,q} := \bigoplus_{m_1+\cdots+m_q=-p} \overline{C}_{m_1} \otimes \cdots \otimes \overline{C}_{m_q}$$

and differentials

$$d_H : \mathsf{Cobar}_{p,q} \to \mathsf{Cobar}_{p+1,q}, \quad x_1 \otimes \cdots x_q \mapsto \sum_{i=1}^{q} (-1)^{\sigma(x_i)} x_1 \otimes \cdots \otimes \partial(x_i) \otimes \cdots \otimes x_q$$

$$d_V : \mathsf{Cobar}_{p,q} \to \mathsf{Cobar}_{p,q+1}, \quad x_1 \otimes \cdots \otimes x_q \mapsto \sum_{i=1}^{q} (-1)^{\sigma(x_i)} x_1 \otimes \cdots \otimes \overline{\Delta}(x_i) \otimes \cdots \otimes x_q$$

where $\sigma(x_i)$ is the Koszul sign $\deg x_1 + \cdots + \deg x_{i-1}$.

Warning 4.4. We have overloaded the notation Cobar, first to refer to the functor from Definition 4.1 sending a dg coalgebra to a dg algebra, and second to represent the functor from Definition 4.3 sending a dg coalgebra to a bicomplex. We will distinguish these using subscripts: using $\mathsf{Cobar}_\bullet$ for the first and $\mathsf{Cobar}_{\bullet,\bullet}$ for the second. But the next proposition shows that this distinction is not so important.

Proposition 4.5. *The totalization of* $\mathsf{Cobar}_{\bullet,\bullet}(C)$ *given by*

$$\mathrm{Tot}_k(\mathsf{Cobar}_{\bullet,\bullet}) = \bigoplus_{p+q=-k} \mathsf{Cobar}_{p,q}$$

with differential $\partial_{\mathrm{Tot}} = d_H + (-1)^q d_V$ *on* $\mathsf{Cobar}_{\bullet,q}$*, is isomorphic as a dg algebra to* $\mathsf{Cobar}_\bullet(C)$.

Proof. By Equation 4.2, we have an equality of groups $\mathrm{Tot}_k(\mathsf{Cobar}_\bullet^\bullet)$ and $\mathsf{Cobar}_k(C)$. Since d_H is of degree $(+1, 0)$ and d_V is of degree $(0, +1)$, the total differential $\partial_{\mathsf{Cobar}}$ has total degree -1. By Lemma 2.7, it suffices to show they agree on the component $s^{-1}\overline{C}$. Indeed, under the identification in Equation 4.2, we have

$$(s^{-1}\overline{C})_p = \mathsf{Cobar}_{p,1}$$

and on this component ∂_{Tot} is precisely $d_H - d_V$. □

What is miraculous is that this purely algebraic construction encodes a good amount of topological information. This is the content of Adams' theorem:

Theorem 4.6 ([Ada56])**.** *Let* (X, x) *be a simply connected based topological space. Let* $\Omega_x X$ *be the space of loops in* X *based at* x*. Then there is a natural isomorphism*

$$H_\bullet(\mathsf{Cobar}(C_\bullet(X))) \cong H_\bullet(\Omega_x X).$$

Thus we can regard the cobar construction as the algebraic analogue of the based loop space functor. As the prefix *co-* suggests, the cobar construction is adjoint to a functor

$$\mathsf{Bar} : \mathsf{dgAlg} \to \mathsf{dgConilCoalg}$$

called the *bar construction*, which is (roughly) the algebraic version of the delooping of a space. We comment briefly on the bar construction in the following few sections.

Remark 4.7. The name *bar* construction comes from typesetting. Pre-LaTeX, a tensor $x_1 \otimes \cdots \otimes x_n$ would be written $x_1|\cdots|x_n$. Adams attributes the name *cobar* to H. Cartan (see footnote 3 of [Ada56]).

4.2. **Two-sided version.** The cobar construction of a dg coalgebra is a special case of a more general *two-sided* cobar construction. An analogous two-sided bar construction is well documented (for example in [Zha19], [MZZ25]). Here we give an exposition of the dual construction, which seems to not appear in the literature.

Definition 4.8. Let C be a conilpotent R-coalgebra and let N, M be left and right comodules over C, respectively. The *two-sided cobar construction on* (N, C, M) is the cosimplicial R-module

$$\mathsf{Cobar}(N, C, M)^q = M \otimes C^{\otimes q} \otimes N$$

with cofaces $d^i : \mathsf{Cobar}^{q-1} \to \mathsf{Cobar}^q$ given by

$$d^i = \begin{cases} \Delta^R \otimes \mathrm{id}^{\otimes q} & i = 0 \\ \mathrm{id}^{\otimes i} \otimes \Delta \otimes \mathrm{id}^{q-i} & i \in \{1, \dots, q-1\} \\ \mathrm{id}^{\otimes q} \otimes \Delta^L & i = q \end{cases}$$

and codegeneracies $s^i : \mathsf{Cobar}^{q+1} \to \mathsf{Cobar}^q$ given by

$$s^i = \mathrm{id}^{\otimes i} \otimes \epsilon \otimes \mathrm{id}^{\otimes q-i-1}$$

By the Dold–Kan correspondence we can also regard this as a chain complex. If C is additionally differential graded, and N, M are dg comodules over C, then then we also get an internal differential on $\mathsf{Cobar}(N, C, M)$ in addition to the differential coming from the coalgebra and comodule maps. Taking these new differentials into account, we obtain the following notion, generalizing the previous discussion of the cobar bicomplex:

Definition 4.9. Let (C, ∂_C), (N, ∂_N), and (M, ∂_M) be as above. The *two-sided cobar bicomplex* has underlying R-module

$$\mathsf{Cobar}(N, C, M)_{\bullet,\bullet} = \bigoplus_{q \geq 0} M \otimes C^{\otimes q} \otimes N$$

and bigrading

$$\mathsf{Cobar}(N, C, M)_{p,q} = \bigoplus_{a+b_1+\cdots+b_q+c=-p} M_a \otimes C_{b_1} \otimes \cdots \otimes C_{b_q} \otimes N_c$$

with vertical differential given by the alternating sums of the coface maps

$$d_V : \mathsf{Cobar}(N, C, M)_{\bullet,q-1} \to \mathsf{Cobar}(N, C, M)_{\bullet\bullet}, q, \quad d_V = \sum_{i=0}^{q} d^i$$

and horizontal differential coming from the comodule and coalgebra differentials:

$$\begin{aligned} d_H : \mathsf{Cobar}(N, C, M)_{p,q} &\to \mathsf{Cobar}(N, C, M)p-1, q \\ d_H(m \otimes c_1 \otimes \cdots \otimes c_q \otimes n) &= \partial_M(m) \otimes c_1 \otimes \cdots \otimes c_q \otimes n \\ &\quad + \sum_{i=1}^{q} (-1)^{|m|+\sigma(c_i)} m \otimes c_1 \otimes \cdots \otimes \partial_C(c_i) \otimes \cdots \otimes c_q \otimes n \\ &\quad + (-1)^{|m|+\sigma(c_q)} m \otimes c_1 \otimes \cdots \otimes c_q \otimes \partial_N(n) \end{aligned} \tag{4.10}$$

where $\sigma(x_k) = |x_1| + \cdots |x_{k-1}|$ is the Koszul sign. Then totalizing with total degree $-p-q$ and differential $d_H + (-1)^q d_V$ on $\mathsf{Cobar}(N, C, M)_{\bullet,q}$ yields a dg algebra, the *two-sided cobar construction* of (N, C, M).

Once again we are overloading the notation Cobar, but having multiple arguments in $\mathsf{Cobar}(-)$ indicates that we are refering to the two-sided version. Of course, this two-sided construction reduces to the one in the previous subsection, which we will sometimes refer to by the *single-sided* cobar construction. To prove this fact, we need a few lemmas and definitions.

Lemma 4.11. *Let $f : D_{\bullet,\bullet} \to E_{\bullet,\bullet}$ be a map of second quadrant bicomplexes of abelian groups with differentials of bidegree $(+1, 0)$ and $(0, +1)$. Then*

(1) If f is a column-wise quasi-isomorphism, then the induced map on totalizations $f_* : \operatorname{Tot} D \to \operatorname{Tot} E$ is a quasi-isomorphism.

(2) If f is a row-wise quasi-isomorphism, and there exists $N \geq 0$ such that $D_{p,q} = E_{p,q} = 0$ for all $q \geq N$, then the induced map on totalizations $f_* : \operatorname{Tot} D \to \operatorname{Tot} E$ is a quasi-isomorphism.

Proof. (1) Let $D^{\leq m}_{\bullet,\bullet}$ be the subcomplex of D where the first m columns (i.e. $p \in \{0, \ldots, -m\}$) are nonzero, and likewise for E. We proceed by induction on m. If $m = 0$ we are done, so suppose for $m \geq 1$ that

$$f_* : \operatorname{Tot}(D^{\leq m-1}) \to \operatorname{Tot}(E^{\leq m-1})$$

is a quasi-isomorphism.

Totalizing by taking total degree $k = -p - q$, we have the short exact sequence

$$0 \to \operatorname{Tot}(F^{\leq m-1} D) \hookrightarrow \operatorname{Tot}(D^{\leq m}) \twoheadrightarrow D_{-m,\bullet}[m] \to 0$$

where $D_{-m,\bullet}[m]$ is the shifted complex

$$D_{-m,0} \to D_{-m,1} \to \cdots$$

where the $D_{-m,0}$ term is in degree m. Then by the five lemma applied to the following diagram and the corresponding diagram in the long exact sequence on homology

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Tot}(D^{\leq m-1}) \arrow[r, hook] \arrow[d] & \operatorname{Tot}(D^{\leq m}) \arrow[r, two heads] \arrow[d] & D_{-m,\bullet}[m] \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \operatorname{Tot}(E^{\leq m-1}) \arrow[r, hook] & \operatorname{Tot}(E^{\leq m}) \arrow[r, two heads] & E_{-m,\bullet}[m] \arrow[r] & 0
\end{tikzcd}$$

we conclude that f_* is a quasi-isomorphism. Now D is a filtered colimit of the $D^{\leq m}$, and homology and totalization commute with filtered colimits, so we conclude that f_* is a quasi-isomorphism $\operatorname{Tot}(D) \to \operatorname{Tot}(E)$.

(2) We equip $\operatorname{Tot} D$ with a filtration inherited from the descending filtration on rows of D and E:

$$F^r \operatorname{Tot}_k D = \bigoplus_{p+q=-k, q \geq r} D_{p,q}$$

and likewise for E. By assumption, this filtration is exhaustive and bounded below, so its associated spectral sequence converges to the homology of the totalization. The first page of the spectral sequence is given by the horizontal homology

$$D^1_{p,q} = H_p(D_{\bullet,q}), \quad E^1_{p,q} = H_p(E_{\bullet,q})$$

and by assumption f induces an isomorphism on this page. Then by the mapping lemma (5.2.4, [Wei95]), f induces an isomorphism on the ∞-pages of D and E. Then applying the five lemma inductively on the finite filtrations of $H_\bullet(\operatorname{Tot} D)$ and $H_\bullet(\operatorname{Tot} E)$ completes the argument.

□

Because we will use this notion many more times, we make the following definition:

Definition 4.12. A map $f : D \to E$ of second-quadrant bicomplexes is a *quasi-isomorphism* if it is a column-wise quasi-isomorphism. (So Lemma 4.11 says that a quasi-isomorphism of second quadrant bicomplexes induces a quasi-isomorphism on totalizations).

Proposition 4.13. *Let C be a conilpotent dg R-coalgebra with coaugmentation $\eta : R \to C$. Set $N = M = R$, regarded as a dg R-comodule with R concentrated in degree zero. Define the left action $\Delta^L : N \to C \otimes N$ via*

$$\begin{tikzcd} R \arrow[d, "\eta"'] \arrow[dr, dashed, "\Delta^L"] & \\ C \arrow[r, "\cong"'] & C \otimes R \end{tikzcd}$$

and likewise for the right action. Then $\mathrm{Tot}_\bullet \mathsf{Cobar}(R, C, R)$ *as above is quasi-isomorphic to the single cobar construction* $\mathsf{Cobar}_\bullet(C)$ *of Definition 4.1.*

Proof. We take the normalized cochain complex on each column $\mathsf{Cobar}_{p,\bullet}$ to obtain:

$$\begin{aligned} N\,\mathsf{Cobar}_{p,q}(R, C, R) &\cong \bigcap_{i=0}^{q-1} \ker(s^i : \mathsf{Cobar}_{p,q} \to \mathsf{Cobar}_{p,q-1}) \\ &= \bigcap_{i=0}^{q-1} \ker(\mathrm{id}^i \otimes \epsilon \otimes \mathrm{id}^{\otimes q-i-1})_p \\ &= (\ker(\epsilon)^{\otimes q})_p \\ &= \bigoplus_{m_1+\cdots+m_q=p} \overline{C_{n_1}} \otimes \cdots \otimes \overline{C_{n_q}} \\ &= \mathsf{Cobar}_{p,q}(C). \end{aligned} \tag{4.14}$$

Upon checking differentials, we conclude that the normalized two-sided cobar bicomplex $N\,\mathsf{Cobar}$ is isomorphic to the one-sided cobar bicomplex on C. Since the inclusion of the normalized complex is a graded quasi-isomorphism, we conclude by Lemma 4.11. □

We hope for the two-sided cobar construction to be functorial enough, i.e. it should send most quasi-isomorphisms of dga coalgebras to quasi–isomorphisms of bicomplexes. As stated, this is not always true, not even for the single-sided cobar construction. An example may be found in Proposition 2.4.3, [LV12]. Regardless, it is possible to state this result if we assume a stronger hypothesis. For the next definition and lemmas, let C, C' be conilpotent dg R-coalgebras with the coradical filtration $F_\bullet$. Let $\mathrm{gr}_r\, C = F_r C / F_{r-1} C$ be the pieces of the associated graded C and likewise for C'. Therefore, any map $f : C \to C'$ induces a map

$$[f] : \mathrm{gr}\, C \to \mathrm{gr}\, C'$$

of associated gradeds.

Definition 4.15. If $[f]$ is a quasi-isomorphism, we say f is a *graded isomorphism*.

The example to keep in mind is the inclusion of the normalized cochain complex into the Moore complex, as it will come up later.

Lemma 4.16. *Let C be a conilpotent dg coalgebra. The inclusion of the normalized cochain complex into the Moore complex is a graded quasi-isomorphism.*

Proof. First note that any coalgebra map $f : C \to C'$ of conilpotent coalgebras preserves the coradical filtration $F_\bullet$. To see this, take an element

$$x \in F_r C = \{x \in C : \forall n \geq r, \overline{\Delta}^n(x) = 0\}.$$

Then, since f commutes with coproducts, we have

$$\overline{\Delta}^n \circ f(x) = (f \otimes \cdots \otimes f) \circ \Delta^n f(x) = 0$$

as desired. Since the chain homotopy equivalence between NC and MC is a linear combination of coalgebra maps (see again the discussion after Corollary 3.2, [LV12]), they too respect the coradical filtraiton. Hence the chain homotopy descends to the associated graded. □

Lemma 4.17. *Let R be a principal ideal domain. Let N, M be left and right conilpotent dg C-comodules. Let N', M' be left and right conilpotent dg C'-comodules. If $f : C \to C'$, $g : N \to N'$, and $h : M \to M'$ are graded quasi-isomorphisms, and C, C', N, N', M, M' and their associated gradeds are flat, then the induced map*

$$(g, f, h)_* : \operatorname{Tot} \mathsf{Cobar}(N, C, M) \to \operatorname{Tot} \mathsf{Cobar}(N', C', M')$$

is a quasi-isomorphism of bicomplexes.

Proof. We imitate the strategy of the proof of the analogous Proposition 2.4.1 in [LV12], which is over a field and done for the regular cobar construction.

Define a weighting on the elements of C, N, M via

$$|x| = \{\min r : x \in F_r\}$$

where F_r is the coradical filtration on the respective modules. Consider the filtration on the two-sided cobar construction given by

$$F_r \mathsf{Cobar}(N, C, M) := \{m \otimes c_1 \otimes \cdots \otimes c_q \otimes n : |m| + |c_1| + \cdots + |c_q| + |n| \leq r\}.$$

The totalization of Cobar inherits this filtration:

$$F_r \operatorname{Tot}_k \mathsf{Cobar}(N, C, M) := \bigoplus_{p+q=-k} F_r \mathsf{Cobar}_{p,q}.$$

The filtration is bounded below and exhaustive by conilpotency, so its associated spectral sequence converges to the homology of $\operatorname{Tot} \mathsf{Cobar}(N, C, M)$. The zeroth page of this spectral sequence is given by

$$E^0_{s,t} \mathsf{Cobar}(N, C, M) = \frac{F_s \operatorname{Tot}_{s+t} \mathsf{Cobar}}{F_{s-1} \operatorname{Tot}_{s+t} \mathsf{Cobar}} = \bigoplus_{q \geq 0} \frac{F_s (M \otimes C^{\otimes q} \otimes N)_{s+t}}{F_{s-1}(M \otimes C^{\otimes q} \otimes N)_{s+t}}.$$

Because all the modules are flat, we can commute the associated graded with the tensor products to obtain

$$E^0_{s,t} \mathsf{Cobar}(N, C, M) = \bigoplus_{q \geq 0} \bigoplus_{a+b_1+\cdots+b_q+c=s} (\operatorname{gr}_a M \otimes \operatorname{gr}_{b_1} C \otimes \cdots \otimes \operatorname{gr}_{b_q} \otimes \operatorname{gr}_c N)_{s+t}.$$

Summing over t, each column of the E^0 page is therefore given by

$$E^0_{s,\bullet} = \mathsf{Cobar}(\operatorname{gr} N, \operatorname{gr} C, \operatorname{gr} M)^{(s)}$$

where the superscript (s) denotes the subcomplex of elements satisfying

$$|n| + |c_1| + \otimes + |c_q| + m = s.$$

In other words, the induced map

$$E^0(\mathsf{Cobar}(g, f, h)) : E^0 \mathsf{Cobar}(\operatorname{gr} N, \operatorname{gr} C, \operatorname{gr} M) \to E^0 \mathsf{Cobar}(\operatorname{gr} N, \operatorname{gr} C', \operatorname{gr} M)$$

is exactly the induced map $\mathsf{Cobar}([g], [f], [h])$. So it suffices to prove that the column-wise maps are quasi-isomorphisms in light of Lemma 4.11.

To do so, we define another filtration G on each column $E^0_{s,\bullet}$ based on tensor length, with

$$n \otimes c_1 \otimes \cdots \otimes c_q \otimes m \in G_k E^0_{s,\bullet} \iff q \geq -k.$$

This is an exhaustive increasing filtration and is bounded below, so the associated spectral sequence, which we will denote by $\tilde{E}$, converges. The filtration G_k/G_{k-1} isolates elements with tensor length exactly k. Because the vertical differentials induced by the coproduct increase tensor length, they are

zero on $\tilde{E}^0$, so the d^0 differential solely comes from the internal differentials of $\operatorname{gr} N, \operatorname{gr} C, \operatorname{gr} M$, i.e. the $\tilde{E}^0$-page is just the tensor product of the chain complexes $\operatorname{gr} N \otimes (\operatorname{gr} C)^{\otimes q} \otimes \operatorname{gr} M$. By flatness and the Künneth formula, we obtain

$$\tilde{E}^1(C) = H_\bullet(\operatorname{gr} N) \otimes H_\bullet(\operatorname{gr} C)^{\otimes q} \otimes H_\bullet(\operatorname{gr} M).$$

By assumption, $([g],[f],[h])$ induces an isomorphism

$$\tilde{E}^1(C) \to \tilde{E}^1(C')$$

and consequently on the columns of the original bicomplex $E^0_{s,\bullet}$, completing the proof. □

4.3. **Homology and the fundamental group.** Adams' theorem allows one to use homology and the cobar construction to approximate the fundamental group:

Theorem 4.18 ([Sta75], Theorem 5.4)**.** *Let R be a ring and (X,x) a path-connected space. Let $\mathcal{I}$ be the augmentation ideal of the group ring $R[\pi_1(X,x)]$. Let E be the spectral sequence associated to the cobar construction on $C_\bullet(X)$ with basepoint x, and the filtration by tensor length. Then there is an isomorphism*

$$\mathcal{I}^p/\mathcal{I}^{p+1} \cong E^\infty_{-p,p}.$$

A dual version of this theorem with cohomology and the bar construction can be found in [Gad23], Theorem 1.2(3). The goal of this subsection will be to prove the following integral version of this theorem for the two-sided cobar construction:

Theorem 4.19. *Let X be a path-connected space with basepoints a, b. Let $\mathcal{I}$ be the augmentation ideal of the group ring $\mathbf{Z}\pi_1(X,a)$. Let E be the spectral sequence associated to the two-sided cobar construction*

$$\mathsf{Cobar}(C_\bullet(\{b\}), C_\bullet(X), C_\bullet(\{a\}))$$

where the filtration is given by tensor length. Then there is an isomorphism

$$\mathbf{Z}\pi_X(a,b)\mathcal{I}^p/\mathbf{Z}\pi_X(a,b)\mathcal{I}^{p+1} \to E^\infty_{-p,p}.$$

We will imitate Stallings' strategy. He relies on the following result of Brown:

Theorem 4.20 (Theorem 10.2, [Bro59])**.** *Let C be a dga R-coalgebra. Suppose $C_0 = R$, $H_1(C) = 0$, and (). Then there is a chain equivalence*

$$\mathsf{Cobar}(\mathsf{Bar}(C)) \to C.$$

In our case we specialize to the case where C is the acyclic cochain complex consisting of the integral group ring $\mathbf{Z}G$ concentrated in degree zero. This is useful for the following reason.

Proposition 4.21. *There is a chain equivalence*

$$\mathsf{Cobar}(C_\bullet(BG)) \to \mathbf{Z}G$$

and consequently for any path-connected space X with a single 0-cell a, there is an isomorphism

$$H_0\,\mathsf{Cobar}(C_\bullet(X)) \to \mathbf{Z}\pi_1(X,a).$$

Proof. The bar construction on $\mathbf{Z}G$ is equivalent to singular chains on the Eilenberg–MacLane space $K(G,1)$. This is an intermediate step and we don't discuss the bar construction again, so we refer the reader to sources such as [Mil67], Chapter 16 of [Bro82], or Chapter 1.5 of [May99].

The second isomorphism follows by Corollary 5.3 of [Sta75], which says that the map of X into the $K(\pi_1(X,a),1)$ induces an isomorphism on cobar constructions. □

Stallings' proof then proceeds via the observation that that under this isomorphism, the tensor length truncations of Cobar correspond to the $\mathcal{I}$-adic filtration on the fundamental group ring. In order to generalize Stallings' theorem, we make the following constructions. Recall that the *fundamental groupoid* of a space X is the category Π_X whose objects are points in X and morphisms are homotopy classes of paths between two points.

Definition 4.22. $\mathbf{Z}\Pi_X$ is the category whose objects are points in X and morphisms are the free abelian group on homotopy classes of paths.

Definition 4.23. The *cobar category* of a space X is the category $\mathfrak{C}\mathsf{obar}(X)$ whose objects are the zero-simplices of X and morphisms between $a, b \in X^0$ is zeroth homology of the two-sided cobar construction

$$H_0 \operatorname{Tot}_\bullet \mathsf{Cobar}(C_\bullet(\{b\}), C_\bullet(X), C_\bullet(\{a\})).$$

Both categories above are enriched in abelian groups. Notice that when X has only a single zero simplex, then $\mathfrak{C}\mathsf{obar}(X)$ is a one-object category whose hom-set the zeroth homology of the cobar construction on $C_\bullet(X)$ by Proposition 4.13. In this case, Prop 4.21 implies that we have a fully faithful functor

$$\mathfrak{C}\mathsf{obar}(X) \to \mathbf{Z}\Pi(X).$$

This generalizes to the multiple basepoint case:

Proposition 4.24. *If X is path-connected then there is a fully faithful functor $\mathfrak{C}\mathsf{obar}(X) \to \mathbf{Z}\Pi(X)$.*

Proof. We define the functor F to be the identity on objects. For $a, b \in X^0$ we have to show that there is an isomorphism

$$F_{a,b} : H_0 \operatorname{Tot} \mathsf{Cobar}(C_\bullet(\{b\}), C_\bullet(X), C_\bullet(\{a\})) \to \mathbf{Z}\Pi_X(a, b).$$

Consider first the map $F'_{a,b} : \ker(\partial_{\mathsf{Cobar}}) \to \mathbf{Z}\Pi_X(a, b)$ defined via

$$F'_{a,b}(1_b \otimes c_1 \otimes \cdots \otimes c_q \otimes 1_a) = 1_b \cdot (\prod_{i=1}^{q}(1 - [c_i])) \cdot 1_a$$

for a cycle $1_b \otimes c_1 \otimes \cdots \otimes c_q \otimes 1_a$. Note that the multiplication in $\mathbf{Z}\pi_X(a, b)$ between two paths is given by concatenation if they share endpoints, and zero otherwise. To see this descends to H_0 of cobar, let $\sigma \in C_2(X)$ be a 2-simplex. By definition of the two-sided cobar differential, we have

$$\begin{aligned}
F'(\partial\sigma) &= (1 - [d_0\sigma]) - (1 - [d_1\sigma]) + (1 - [d_2\sigma]) \\
&= 1 - [d_0\sigma] + [d_1\sigma] - [d_2\sigma] \\
F'(\overline{\Delta}\sigma) &= (1 - [d_2\sigma])(1 - [d_0\sigma]) \\
&= 1 - [d_0\sigma] - [d_2\sigma] + [d_2\sigma * d_0\sigma] \\
F' \circ \partial_{\mathsf{Cobar}}(1_a \otimes \sigma \otimes 1_b) &= F'(\partial\sigma - \overline{\Delta}(\sigma)) \\
&= [d_1\sigma] - [d_2\sigma * d_0\sigma] = 0
\end{aligned} \tag{4.25}$$

since $d_2\sigma * d_0\sigma$ is homotopic to $d_1\sigma$. Thus $F'_{a,b}$ descends to a well defined map

$$F_{a,b} : H_0 \operatorname{Tot} \mathsf{Cobar}(C_\bullet(\{b\}), C_\bullet(X), C_\bullet(\{a\})) \to \mathbf{Z}\Pi_X(a, b).$$

To see it is an isomorphism, note that by choosing a path σ from a to b, we can make $\Pi_X(a, b)$ into a $\pi_1(X, a)$-torsor, which induces an isomorphism on the free abelian groups

$$\mathbf{Z}\pi_1(X, a) \to \mathbf{Z}\pi_X(a, b)$$

given by concatenation with σ. Similarly, we have a isomorphism

$$\mathsf{Cobar}(C_\bullet(X)) \cong \mathsf{Cobar}(C_\bullet(\{a\}), C_\bullet(X), C_\bullet(\{a\})) \to \mathsf{Cobar}(C_\bullet(\{b\}), C_\bullet(X), C_\bullet(\{a\}))$$

induced by the isomorphism $C_\bullet(\{a\}) \to C_\bullet(\{b\})$ coming from concatenation with σ. These isomorphisms fit into a commutative diagram

$$\begin{tikzcd} H_0 \operatorname{Cobar}(C_\bullet(X)) \arrow[r, "\cong, 4.21"] \arrow[d, "\cong"'] & \mathbf{Z}\pi_1(X,a) \arrow[d, "\cong"] \\ H_0 \operatorname{Cobar}(C_\bullet(b), C_\bullet(X), C_\bullet(a)) \arrow[r, dashed, "F_{a,b}"'] & \mathbf{Z}\Pi_X(a,b) \end{tikzcd}$$

implying that $F_{a,b}$ too is an isomorphism. □

Theorem 4.19 then follows immediately by noticing that under $F_{a,b}$, the tensor-length filtration on the two-sided cobar construction corresponds to the $\mathcal{I}$-adic filtration on $\mathbf{Z}\pi_X(a,b)$. Thus the E^∞-page computes the associated graded of $\mathbf{Z}\pi_X(a,b)$ under this filtration.

5. Proof of main theorem

Here we present the proof of Theorem 1.1. Throughout this section let X be a path connected topological space with basepoints a, b. Our strategy will be to consider the spectral sequence associated to the two-sided cobar construction. We will show that it converges to the group $H_n(X^n, X(n)^a_b)$ and conclude by the previous discussion relating the cobar construction to the fundamental group.

5.1. **Building the spectral sequence.** Recall the cosimplicial path space $P^\bullet_{a,b}X$ of Example 3.3. From now we denote it by $P^\bullet$ for simplicity. Notice first that the union of the images of the coface maps $d^i : X^{n-1} \to X^n$ is exactly the subspace $X(n)^a_b$. This is a good sign.

Now consider the bicomplex $\mathscr{C}_{\bullet,\bullet}$ which is obtained by applying singular chains to $P^\bullet$:

$$\mathscr{C}_{p,q} = C_{-p}(P^q)$$

with "horizontal" differential $d^{\mathscr{C}}_H : C_p(P^q) \to C_{p+1}(P^q)$ given by the singular boundary map, and "vertical" differential $d^V_{\mathscr{C}} : C_p(P^q) \to C_p(P^{q+1})$ given by the differential of the Moore complex of the cosimplicial abelian group $C_p(P^\bullet)$.

Each column p of this bicomplex forms a cochain complex $\mathscr{C}_{p,\bullet}$, the Moore complex of $C_p(F^\bullet)$. Consider now the bicomplex $\mathscr{N}_{\bullet,\bullet}$ whose columns are instead the normalized cochain complexes of $C_p(F^\bullet)$. The map $\mathscr{C} \twoheadrightarrow \mathscr{N}$ is a quasi-isomorphism of bicomplexes by Dold–Kan and Lemma 4.11. Let us spell out what the groups in $\mathscr{N}$ actually are. Following the definition, we have

$$\mathscr{N}_{p,q} = \operatorname{coker} \bigoplus_{i=1}^{n} ((d^i)_* : C_p(X^{n-1}) \to C^p(X^n)) = C^p(X^n)/\langle \sum_{i=1}^{n} \operatorname{im}(d^i)_* \rangle = C_p(X^n, X(n)_b).$$

Moreover, the vertical differential $\mathscr{N}_{p,q} \to \mathscr{N}_{p,q+1}$ is induced by the zeroth coface map d^0. Consider now the bicomplex obtained from $\mathscr{N}$ by setting all rows above $q = n$ to zero, which we will denote by $(\mathscr{N}^{\leq n})^\bullet_\bullet$.

Proposition 5.1. *Filtering $(\mathscr{N}^{\leq n})$ by the lower index, we obtain a second-quadrant spectral sequence collapsing on the second page with*

$$E^2_{-n,n} = H_n(X^n, X(n)^a_b), \quad E^2_{0,0} = \begin{cases} \mathbf{Z} & a = b \\ 0 & a \neq b \end{cases}$$

converging to $H_\bullet \operatorname{Tot}(\mathscr{N}^{\leq n})$.

Proof. The filtration is bounded below and exhaustive, so the spectral sequence converges. To compute the E^2-page, let us begin on E^0, whose $-p$th column is the following normalized cochain complex (beginning below in degree $q = 0$):

$$C_p(*) \to C_p(X, \{b\}) \to \cdots \to C_p(X^n, X(n)_b) \to 0 \to \cdots$$

of the Moore complex

$$C_p(*) \to C_p(X) \to \cdots \to C_p(X^n) \to 0 \to \cdots \tag{5.2}$$

so they have the same cohomology. But now (5.2) is just the cochain complex obtained by taking the free abelian group on the cosimplicial set $[q] \mapsto \mathrm{Map}(\Delta^p, F^q)$, with terms $q > n$ set to zero. Therefore, by Proposition 3.11, we have that $E^1_{-p,q} = 0$ for $q \neq 0, n$. For $q = n$ and $q = 0$, we have

$$E^1_{-p,n} = C_p(X^n, X(n)_b)/\operatorname{im}(d^0) = C_p(X^n, X(n)^a_b)$$

$$E^1_{-p,0} = \ker(d^0 : C_p(*) \to C_p(X, \{b\})) = \begin{cases} \mathbf{Z} & p = 0, a = b \\ 0 & \text{otherwise} \end{cases}.$$

On the E^1-page, the differential is now induced by singular boundary map. On the first nonzero row $q = n$ of the E^1-page, we just take singular homology to obtain

$$E^2_{-p,n} = H_p(X^n, X(n)^a_b).$$

And on the other possibly nonzero row $q = 0$, we have

$$E^2_{-p,0} = \begin{cases} H_p(\mathbf{Z} \leftarrow 0 \leftarrow \cdots) & a = b \\ 0 & a \neq b \end{cases}$$

as desired.

At this point, because the only nonzero rows are $q = n$ and possibly $q = 0$, the differentials on the pages $E^{\geq 2}$ are all zero, except possibly on page E^{n+1}, where we have a differential d_{n+1} of bidegree $(n+1, -n)$ from $E^{n+1}_{-n-1,n} = H_{n+1}(X^n, X(n)^a_b)$ to $E^{n+1}_{0,0} = \mathbf{Z}$. To see that $d_{n+1} = 0$, consider the map of pairs $f : (X, x) \to (*, *)$. Repeating the constructions in this section, we get a spectral sequence $D_{\bullet,\bullet}$ computing $H_n(*^n, *(n)^*_*) = 0$. By functoriality, f induces a morphism of spectral sequences. Specializing to the $n+1$th page, we have a commutative square

$$\begin{tikzcd}
H_{n+1}(X^n, X(n)^x_x) \arrow[r, equal] & E^{n+1}_{-n-1,n} \arrow[r, "d^E_{n+1}"] \arrow[d, "f_*"'] & E^{n+1}_{0,0} \arrow[r, equal] \arrow[d, "f_*"] & \mathbf{Z} \\
H_{n+1}(*, *) = 0 \arrow[r, equal] & D^{n+1}_{-n-1,n} \arrow[r, "d^D_{n+1}"'] & D^{n+1}_{0,0} \arrow[r, equal] & \mathbf{Z}
\end{tikzcd}$$

But now the $f_* : \mathbf{Z} \to \mathbf{Z}$ is the identity since it is the induced map $H_0(X, x) \to H_0(*, *)$, forcing $d^E_{n+1} = 0$, so $E^2 = E^\infty$ as desired. □

Corollary 5.3. *We have a surjection*

$$H_0(\mathrm{Tot}(\mathscr{N}^{\leq n})) \twoheadrightarrow H_n(X^n, X(n)^a_b)$$

whose kernel is $\mathbf{Z}$ *if* $a = b$*, and trivial otherwise.*

Proof. Follows from convergence of the spectral sequence. □

5.2. **Functoriality.** It remains to compute H_0 of this totalization. For this we will return to our original bicomplex $\mathscr{C}$ and show that its truncated totalization is quasi-isomorphic to the totalization of the two-sided cobar bicomplex. Instead of normalizing the columns, let us first consider the rows $\mathscr{C}_{\bullet,q}$, which are the singular chain complexes $C_\bullet(X^q)$. Then we can apply the topological Alexander–Whitney map on each row:

$$\mathscr{C}_{-\bullet,q} = C_\bullet(X^q) \to C_\bullet(X)^{\otimes q}$$

to obtain a new bicomplex, which we will call $\mathscr{D}$. By definition of tensor product of chain complexes, $\mathscr{D}$ has a bigrading given by

$$\mathscr{D}_{-p,q} = (C_\bullet(X)^{\otimes q})_p = \bigoplus_{n_1+\cdots+n_q=p} C_{n_1}(X) \otimes \cdots \otimes C_{n_q}(X). \tag{5.4}$$

Let $\mathscr{C}^{\leq n}$ and $\mathscr{D}^{\leq n}$ be the bicomplexes obtained from $\mathscr{C}$ and $\mathscr{D}$ by setting all rows of above $q = n$ to zero.

Lemma 5.5. *The map $\mathscr{C}^{\leq n} \to \mathscr{D}^{\leq n}$ induced by Alexander–Whitney is a quasi-isomorphism of bicomplexes, with inverse quasi-isomorphism induced by the Eilenberg–Zilber map.*

Proof. Combine Corollary 3.16, Theorem 3.17, and Lemma 4.11, □

Theorem 5.6. *Let $C_\bullet(-)$ be the singular chains functor. Regard $C_\bullet(\{a\})$ as a right $C_\bullet(X)$-comodule via the right comodule action*

$$\Delta^R : C_\bullet(\{a\}) \xrightarrow{\mathcal{AW}} C_\bullet(\{a\}) \otimes C_\bullet(\{a\}) \xrightarrow{\mathrm{id} \otimes \iota_a} C_\bullet(\{a\}) \otimes C_\bullet(X)$$

and likewise for $C_\bullet(\{b\})$ as a left $C_\bullet(X)$-comodule. Then there is a quasi-isomorphism

$$\mathsf{Cobar}^{\leq n}(C_\bullet(\{b\}), C_\bullet(X), C_\bullet(\{a\})) \to \mathscr{D}^{\leq n}.$$

Proof. Denote $\mathsf{Cobar}(\cdots)$ by Cobar. Singular chains are free abelian groups, hence flat over $\mathbf{Z}$. Moreover, over $\mathbf{Z}$, submodules of free modules are free. Lastly, the associated gradeds of the singular chains are free. To see this, notice that the coradical filtration on singular chains is just the skeletal filtration:

$$F_r C_\bullet(X) = \bigoplus_{k=0}^{r} C_k(X),$$

so that the associated gradeds are just the levelwise singular chain groups, which are free.

Therefore, by Lemma 4.17 and Lemma 4.16, it suffices to exhibit a quasi-isomorphism from

$$N\,\mathsf{Cobar}^{\leq n} := \mathsf{Cobar}^{\leq n}(NC_\bullet(\{b\}), NC_\bullet(X), NC_\bullet(\{a\})) \tag{5.7}$$

to $\mathscr{D}^{\leq n}$. Now by Lemma 4.11, it suffices to exhibit a quasi-isomorphism from (5.7) to $N\mathscr{D}^{\leq n}$, the truncation of the bicomplex obtained by replacing each column of $\mathscr{C}$ with its normalized chains, then applying the Alexander–Whitney map, as in the diagram below:

$$\begin{tikzcd}
C_\bullet(X^q) \arrow[r, "\mathcal{AW}_q"] & C_\bullet(X)^{\otimes q} = \mathscr{D}_{\bullet,q} \\
NC_\bullet(X^q) \arrow[u, hook] \arrow[r, "\mathcal{AW}_q"'] & NC_\bullet(X)^{\otimes q} =: N\mathscr{D}_{\bullet,q} \arrow[u, hook]
\end{tikzcd}$$

Notice that $NC_\bullet(\{a\})$ and $NC_\bullet(\{b\})$ are just the chain complexes with a $\mathbf{Z}$ term concentrated in degree zero, i.e. the identity object in the symmetric monoidal structure on $\mathsf{Ch}_+(\mathsf{Ab})$. Hence we have natural isomorphisms

$$\psi_a : NC_\bullet(\{a\}) \otimes NC_\bullet(X) \to NC_\bullet(X)$$
$$\psi_b : NC_\bullet(X) \otimes NC_\bullet(\{b\}) \to NC_\bullet(X)$$

which combine to give a natural isomorphism $\phi := \psi_a \otimes \mathrm{id}^{\otimes \bullet} \otimes \psi_b$:

$$\begin{aligned} \phi : N\,\mathsf{Cobar}_{-p,q} &= \bigoplus_{n+b_1+\cdots+b_q+m=p} NC_n(\{a\}) \otimes NC_{b_1}(X) \otimes \cdots \otimes NC_{b_q}(X) \otimes NC_m(\{b\}) \\ &\to \bigoplus_{b_1+\cdots+b_q=p} NC_{b_1}(X) \otimes \cdots \otimes NC_{b_q}(X) =: N\mathscr{D}_{-p,q}. \end{aligned} \tag{5.8}$$

We now check that the differentials on $N\,\mathsf{Cobar}$ and $N\mathscr{D}$ are compatible with ϕ. Because all differentials of $NC_\bullet(\{a\})$ and $NC_\bullet(\{b\})$ are zero, the horizontal cobar differential formula 4.10 yields

$$\begin{aligned} \phi(d_H^{N\,\mathsf{Cobar}}(m \otimes c_1 \otimes \cdots \otimes c_q \otimes n)) &= \phi(\sum_{i=1}^{q} (-1)^{\sigma(c_i)} m \otimes c_1 \otimes \cdots \otimes \partial_{NC}(c_i) \otimes \cdots \otimes c_q \otimes n) \\ &= \sum_{i=1}^{q} (-1)^{\sigma(c_i)} \psi_a(m \otimes c_1) \otimes \cdots \otimes \partial_{NC}(c_i) \otimes \cdots \otimes \psi_b(c_q \otimes n) \\ &= d_{N\mathscr{D}}(\phi(m \otimes c_1 \otimes \cdots \otimes c_q \otimes n)). \end{aligned}$$

The vertical differential in both cases is given by the alternating sum of coface maps d^i. We will show that $\phi \circ d^i_{\mathsf{Cobar}} = d^i_{N\mathscr{D}} \circ \phi$ for all i. For $i = 0$, the map $d^i_{N\mathscr{D}} : N\mathscr{D}^q_\bullet \to N\mathscr{D}^{q+1}_\bullet$ is induced by the following inclusion under the topological Alexander–Whitney map:

$$(x_1, \ldots, x_q) \to (a, x_1, \ldots, x_q).$$

Naturality of Alexander–Whitney implies that the following diagram commutes:

$$\begin{tikzcd} NC_\bullet(X^q) \arrow[r, "\cong"] \arrow[d, "\mathcal{AW}_q"'] \arrow[rr, bend left, "NC_\bullet(d^0)"] & NC_\bullet(\{a\} \times X^q) \arrow[r, "NC_\bullet(\iota_a \times \mathrm{id}_{X^q})"] \arrow[d, "\mathcal{AW}_{\{a\},X^q}"'] & NC_\bullet(X^{q+1}) \arrow[d, "\mathcal{AW}_{q+1}"] \\ NC_\bullet(X)^{\otimes q} \arrow[r, "\cong"] \arrow[rr, bend right, "d^0_{\mathscr{D}}"'] & NC_\bullet(\{a\}) \otimes NC_\bullet(X)^{\otimes q} \arrow[r, "NC_\bullet(\iota_a) \otimes \mathrm{id}^{\otimes q}"] & NC_\bullet(X)^{\otimes q+1} \end{tikzcd}$$

where ι_a is the inclusion $\{a\} \hookrightarrow X$. But now we make the identifications

$$NC_\bullet(\{a\}) \otimes NC_\bullet(X)^{\otimes q} \cong NC_\bullet(X)^{\otimes q}, \text{ and } NC_\bullet(\iota_a) = \eta_a$$

to conclude

$$(\eta_a \otimes \mathrm{id}^{\otimes q}) \circ \mathcal{AW}_q = \mathcal{AW}_{q+1} \circ NC_\bullet(d^0)$$

which implies $d^0_{N\mathscr{D}} = \eta_a \otimes \mathrm{id}^{\otimes q}$. (By a slight abuse of notation we use η_a to denote both the constant map at η_a as well as the constant zero simplex at a). Then by naturality of the Alexander–Whitney map,

$$\begin{aligned} \phi(d^0_{\mathsf{Cobar}}(m \otimes c_1 \otimes \cdots \otimes c_q \otimes n)) &= \phi(\Delta^R(m) \otimes c_1 \otimes \cdots \otimes c_q \otimes n) \\ &= \phi\big(((\mathrm{id} \otimes \iota_a) \circ \mathcal{AW}(m)) \otimes c_1 \otimes \cdots \otimes c_q \otimes n\big) \\ &= \phi(m \otimes \eta_a \otimes c_1 \otimes \cdots \otimes c_q \otimes n) \\ &= \psi_a(m \otimes \eta_a) \otimes c_1 \otimes \cdots \otimes \psi_b(c_q \otimes n) \\ &= \eta_a \otimes \psi_a(m \otimes c_1) \otimes \cdots \otimes \psi_b(c_q \otimes n) \\ &= (\eta_a \otimes \mathrm{id}^{\otimes q})\big(\psi_a(m \otimes c_1) \otimes \cdots \otimes \psi_b(c_q \otimes n)\big) \\ &= d^0_{N\mathscr{D}}(\phi(m \otimes c_1 \otimes \cdots \otimes c_q \otimes n)) \end{aligned}$$

The proof for $d^1, \ldots, d^q$ are analogous. Hence the restriction of ϕ to the truncated bicomplexes yields a quasi-isomorphism $N\,\mathsf{Cobar}^{\leq n} \to N\mathscr{D}^{\leq n}$, completing the proof. □

Proof of Theorem 1.1. By Theorem 5.6, Lemma 5.5, and the discussion in the first subchapter, we have a quasi-isomorphism

$$\mathrm{Tot}(\mathsf{Cobar}^{\leq n}) \xrightarrow{5.6} \mathrm{Tot}(\mathscr{D}^{\leq n}) \xrightarrow{5.5} \mathrm{Tot}(\mathscr{C}^{\leq n}) \xrightarrow{chapter} \mathrm{Tot}(\mathscr{N}^{\leq n}).$$

Therefore, on zeroth homology we have an isomorphism

$$H_0\,\mathrm{Tot}\,\mathsf{Cobar}^{\leq n} \to H_0\,\mathrm{Tot}(\mathscr{N}^{\leq n}) \tag{5.9}$$

Now Theorem 4.19 yields an isomorphism

$$H_0\,\mathrm{Tot}(\mathscr{N}^{\leq n}) \cong \mathbf{Z}\pi_X(a,b)/\mathbf{Z}\pi_X(a,b)\mathcal{I}^{n+1}. \tag{5.10}$$

Combining Equations 5.9, 5.10 and applying Corollary 5.3, we obtain a surjection

$$\mathbf{Z}\pi_X(a,b)/\mathbf{Z}\pi_X(a,b)\mathcal{I}^{n+1} \xrightarrow{\cong} H_0\,\mathrm{Tot}\,\mathsf{Cobar}_{\leq n} \xrightarrow{\cong} H_0\,\mathrm{Tot}_\bullet(\mathscr{N}^{\leq n}) \twoheadrightarrow H_n(X^n, X(n)^x_x)$$

whose kernel if isomorphic to $\mathbf{Z}$ is $a = b$ and trivial otherwise. It remains to see that if $a = b$, the kernel is generate by the constant path at a. □

5.3. **Looking forward.**

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